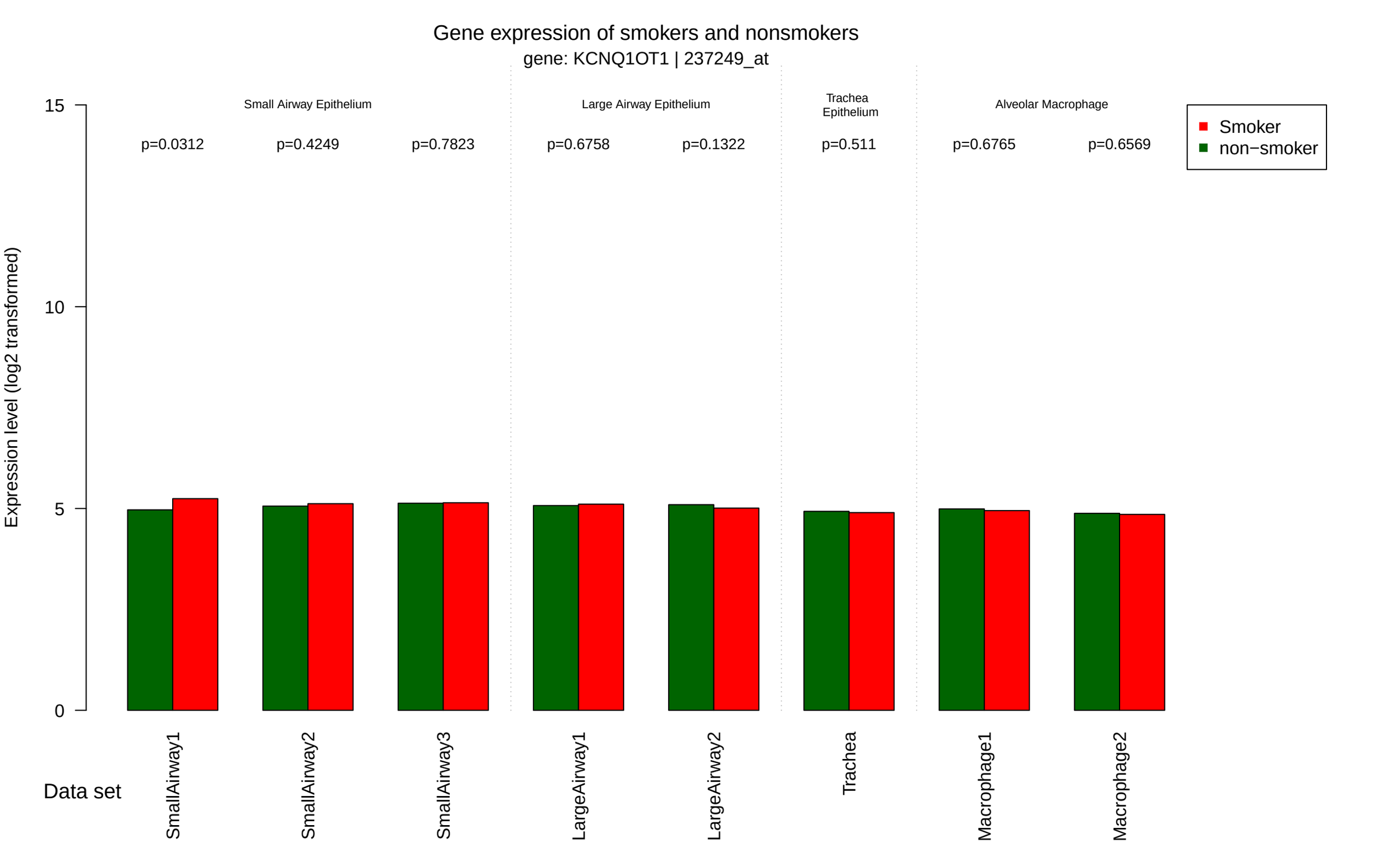
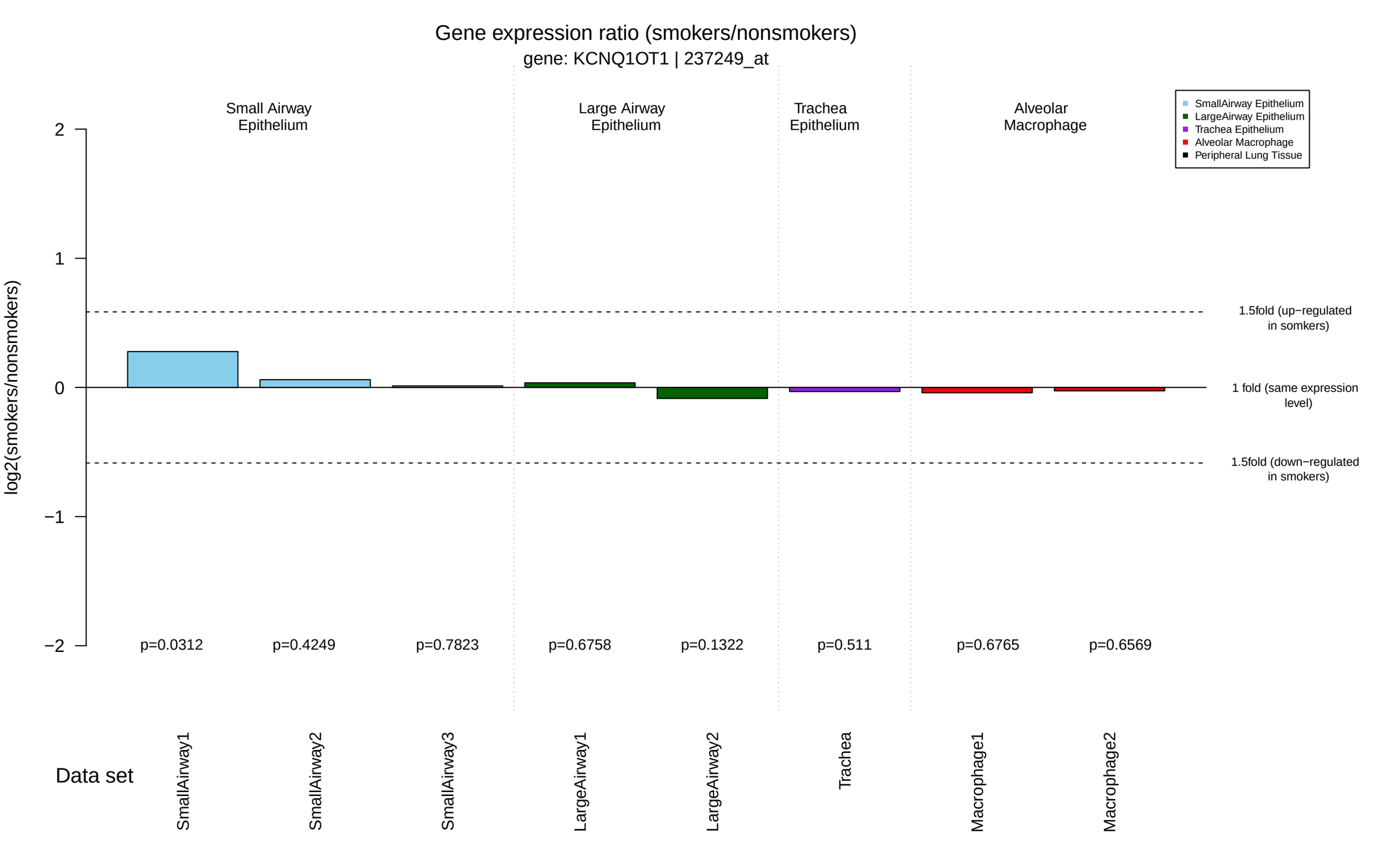
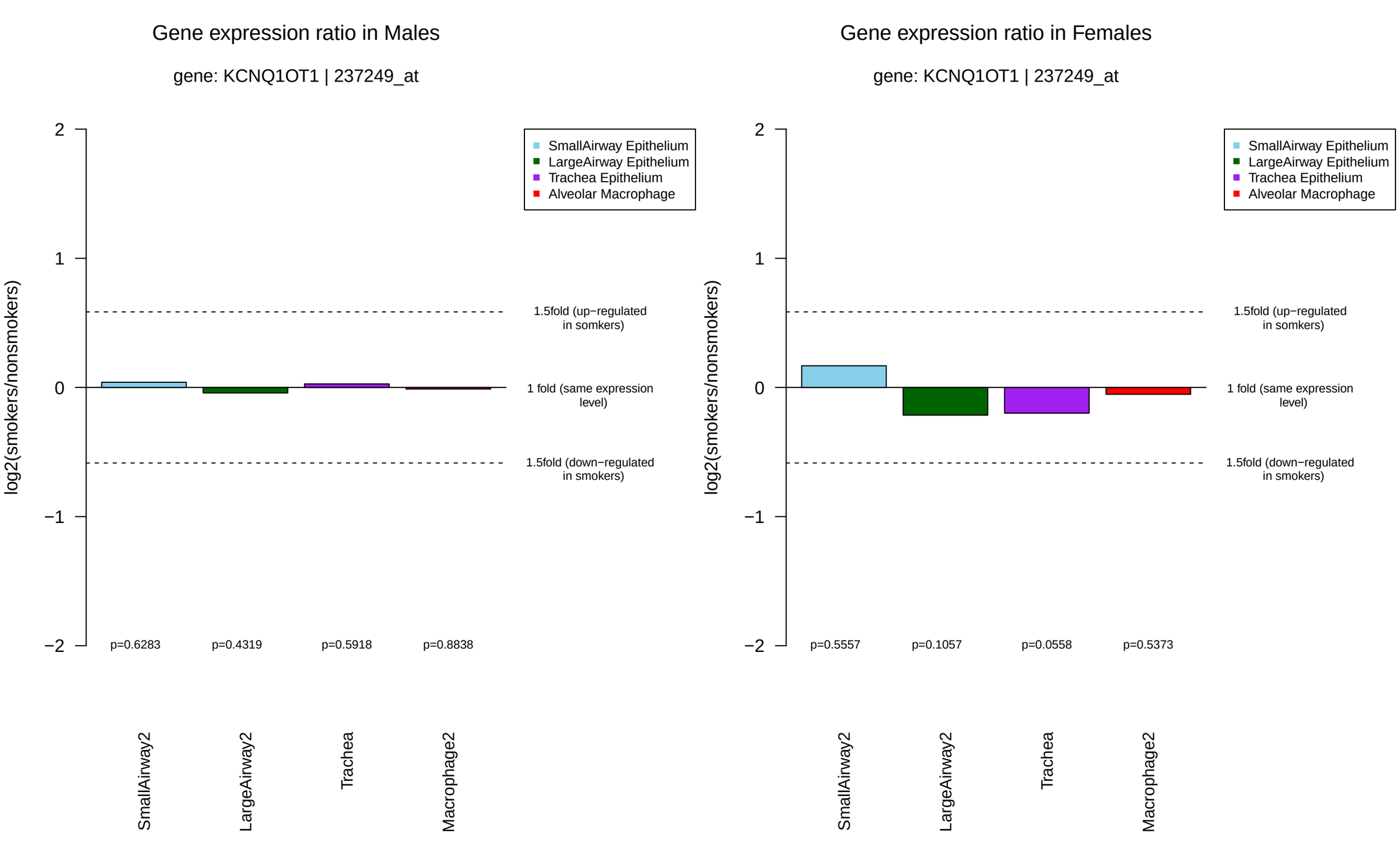
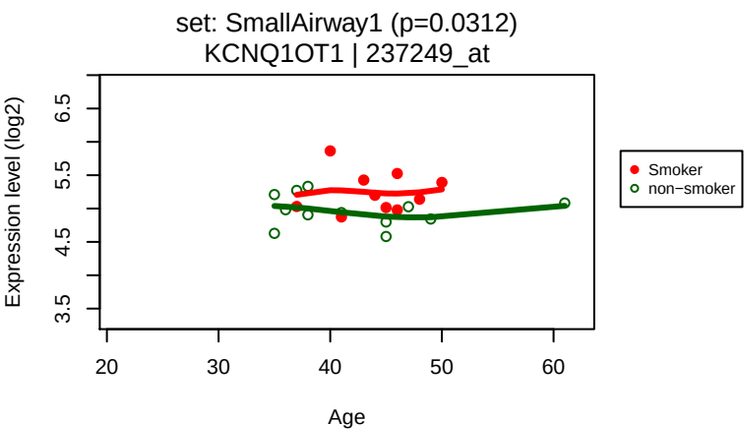
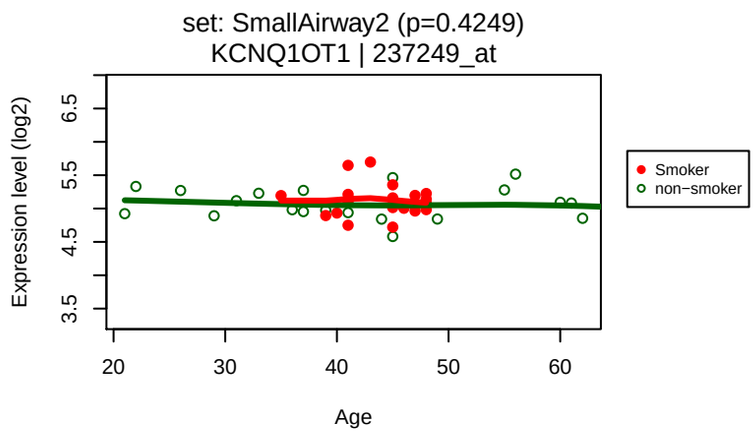
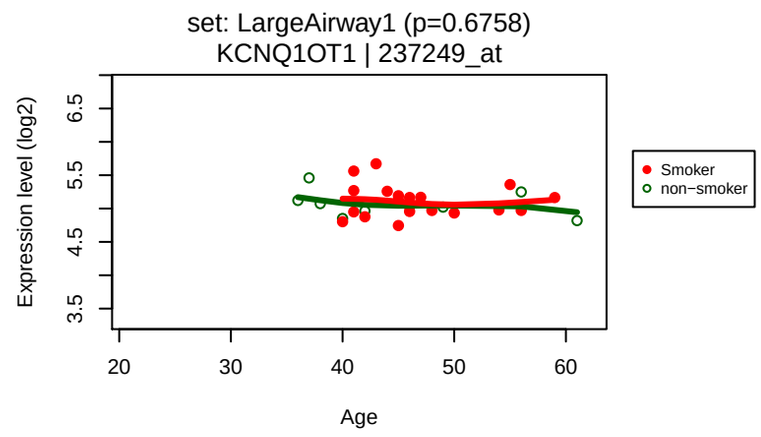

Gene expression of smokers and nonsmokers
gene: KCNQ1OT1 | 237249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0312
p=0.4249
p=0.7823
p=0.6758
p=0.1322
p=0.511
p=0.6765
p=0.6569
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

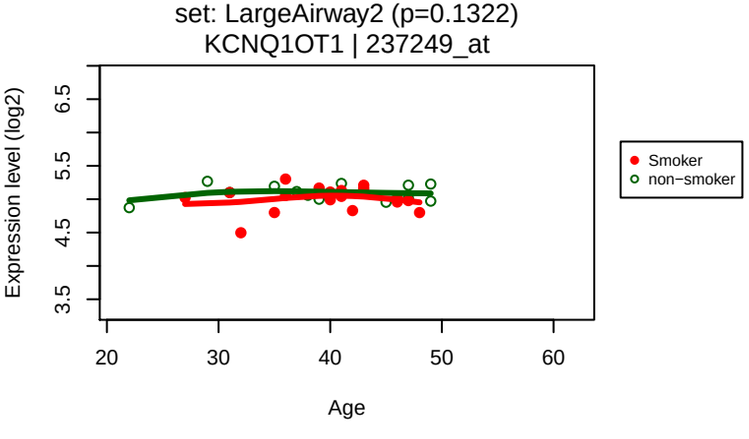

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ1OT1 | 237249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0312
p=0.4249
p=0.7823
p=0.6758
p=0.1322
p=0.511
p=0.6765
p=0.6569
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

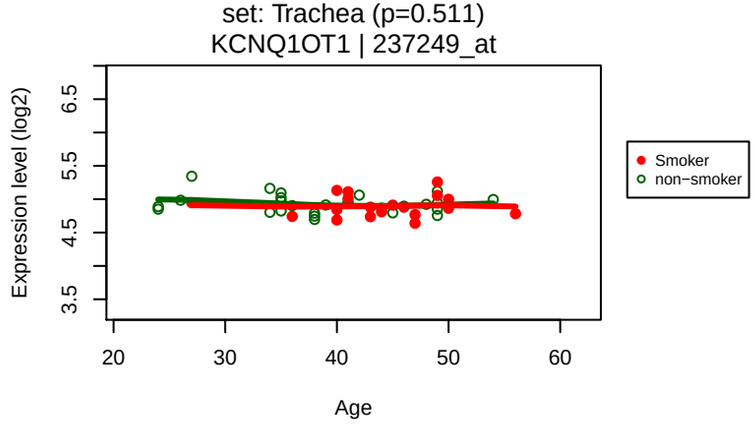

Gene expression ratio in Males
gene: KCNQ1OT1 | 237249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6283
p=0.4319
p=0.5918
p=0.8838
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ1OT1 | 237249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5557
p=0.1057
p=0.0558
p=0.5373
SmallAirway2
LargeAirway2
Trachea
Macrophage2

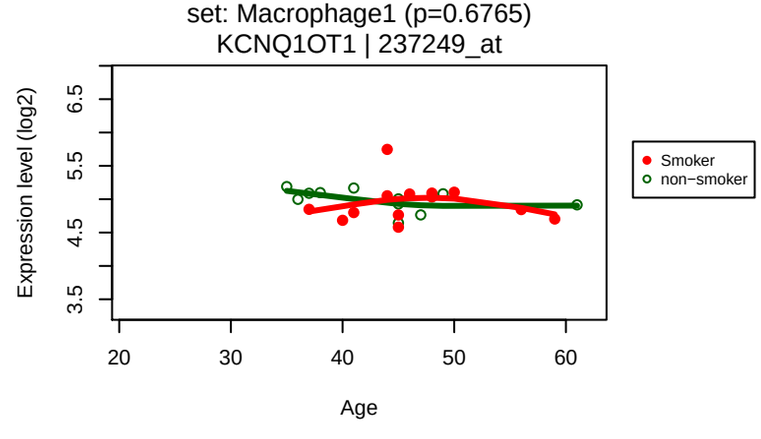

set: SmallAirway1 (p=0.0312)
KCNQ1OT1 | 237249_at
Expression level (log2)
Age
Smoker
non−smoker

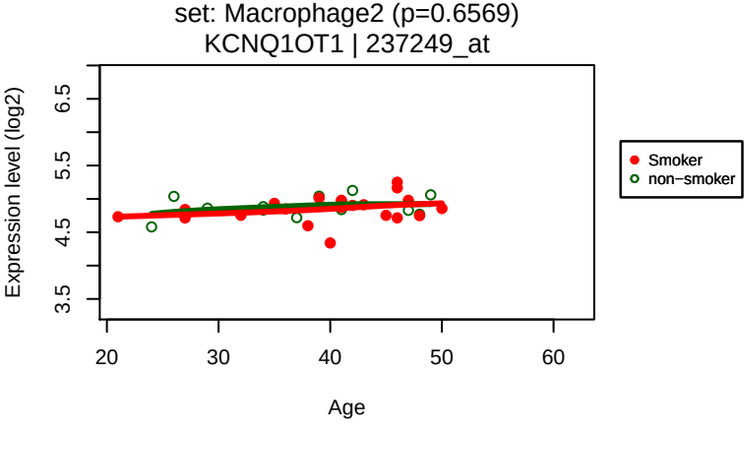

set: SmallAirway2 (p=0.4249)
KCNQ1OT1 | 237249_at
Expression level (log2)
Age
Smoker
non−smoker

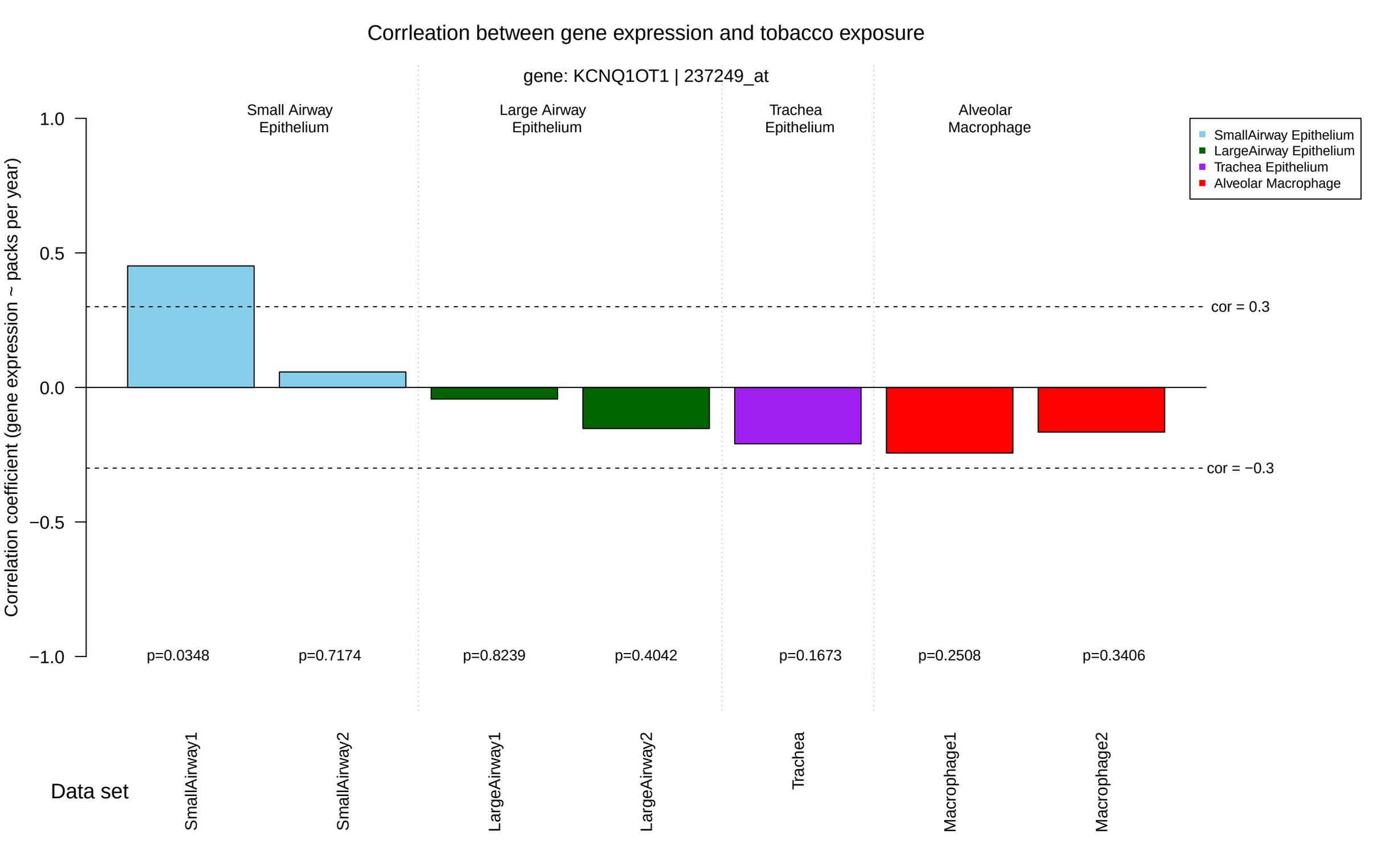

set: LargeAirway1 (p=0.6758)
KCNQ1OT1 | 237249_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1322)
KCNQ1OT1 | 237249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.511)
KCNQ1OT1 | 237249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6765)
KCNQ1OT1 | 237249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6569)
KCNQ1OT1 | 237249_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ1OT1 | 237249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0348
p=0.7174
p=0.8239
p=0.4042
p=0.1673
p=0.2508
p=0.3406
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2